

Single-axis voltage sweep — run report

RUN

230126_sweep_sweep

Status: completed | Started: 2026-09-23T22:01:26.339498+00:00

PURPOSE

Step one output through an explicit RP voltage range; hold the other at zero. Log PAX response. This collects data; it does not fit V_{π} .

REQUIRED OPTICAL SETUP

PAX must observe the selected actuator's optical response; record its location.

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port

Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

OPERATOR SETUP NOTES

V_{π} expectation only; verify channel and measured driver gain before calibration.

DATA

11 recorded samples; acquisition: 11

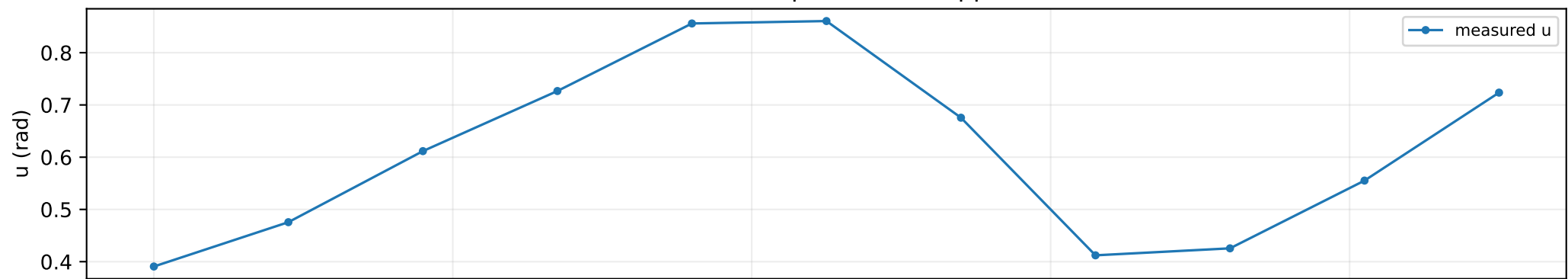
Source: data.csv; full acquisition settings: recipe.json; status: run.json

RAW DOP

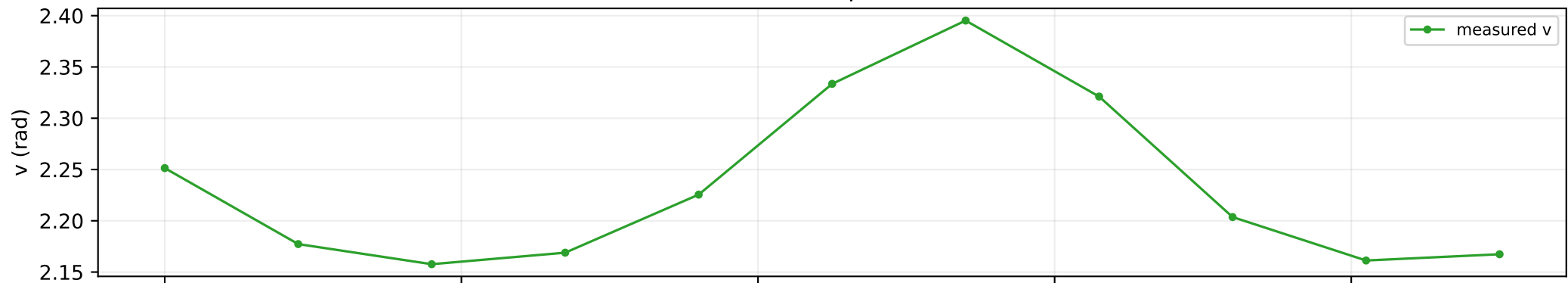
Mean 0.0413; range 0.0331 to 0.0463. 0 samples outside 0-1. Telemetry is not a new calibration.

Calibration sweep — data.csv — swept phi1

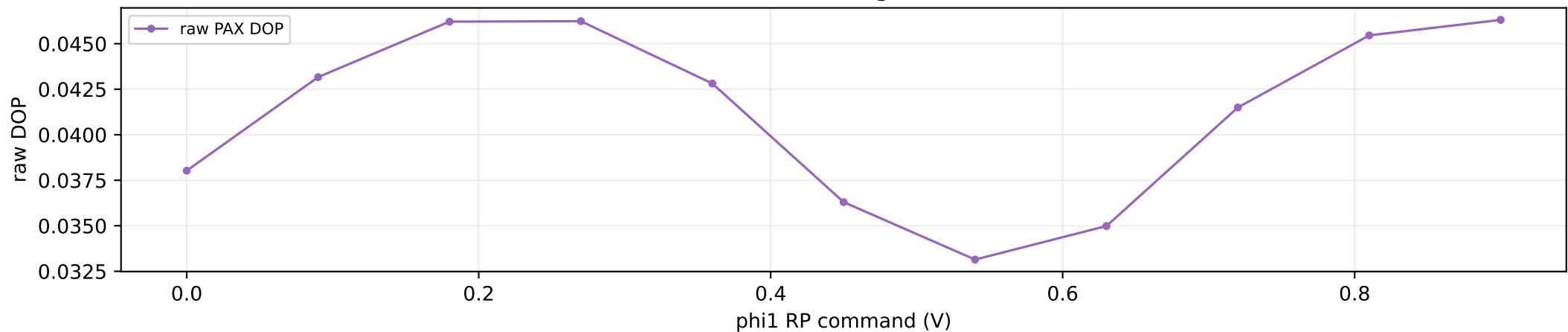
Azimuthal response (unwrapped)



Polar response



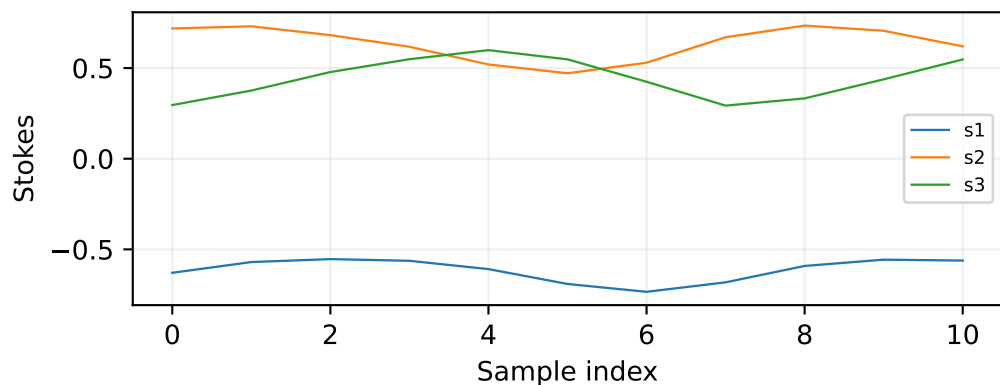
PAX DOP (diagnostic)



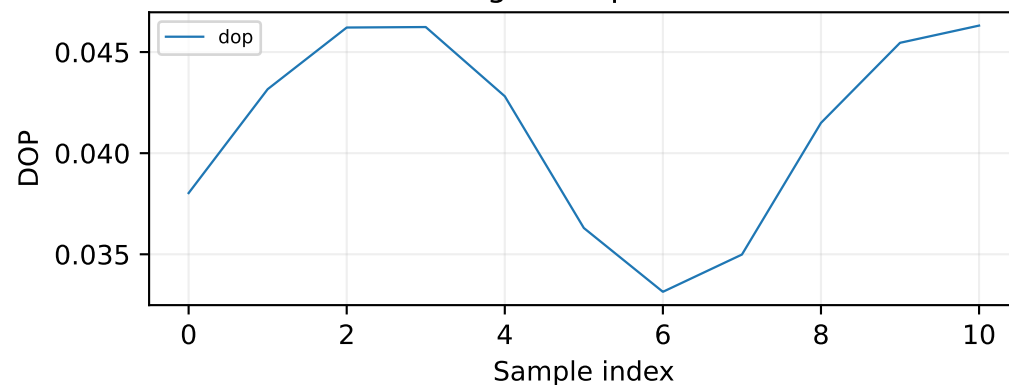
Recorded telemetry

Acquisition order preserves stage changes; no timing alignment or transfer gain is inferred.

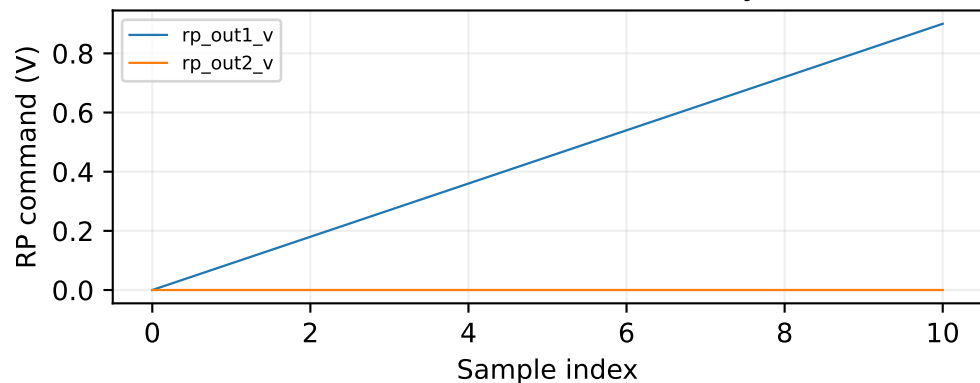
Normalized Stokes



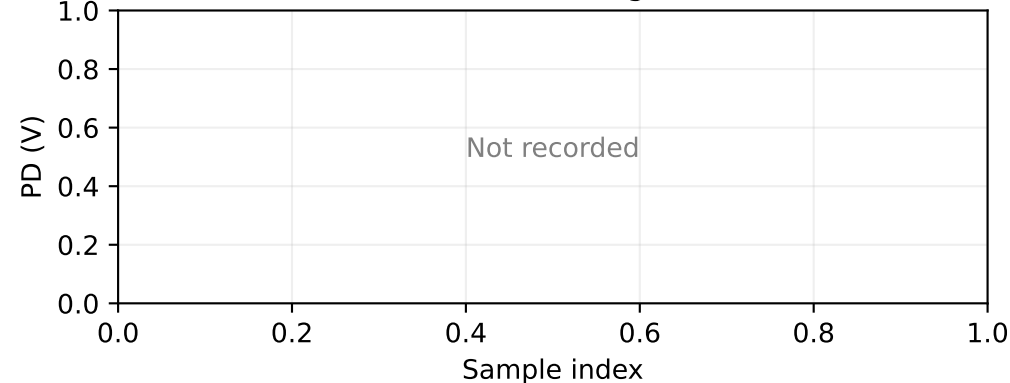
Raw degree of polarization



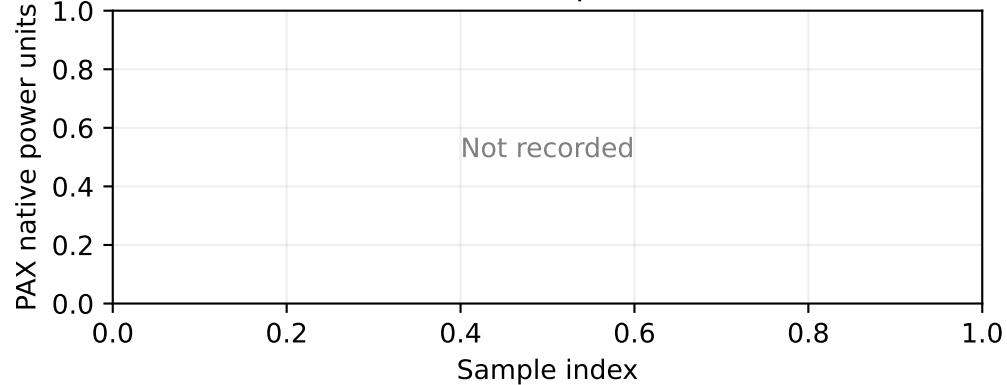
Actuator command history



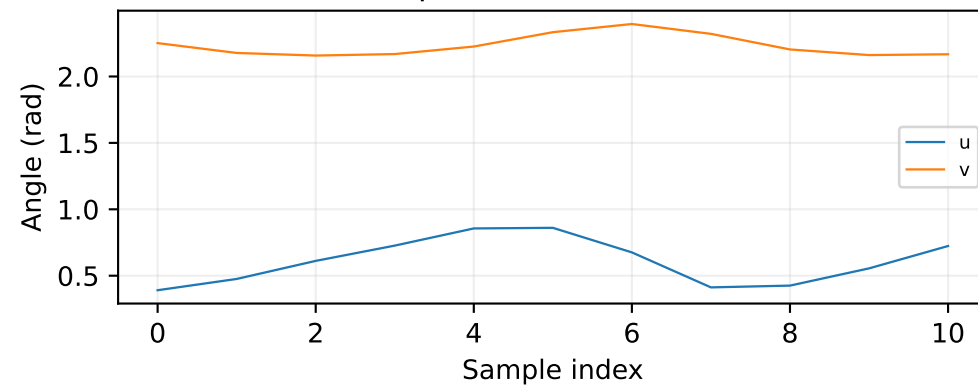
Photodiode signal



PAX total power



Sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π . Gains: terminal V / RP V.

Parameter	Recorded value
label	"sweep"
axis	"phi1"
report	"pdf"
sweep_start_rp_v	0.0
sweep_stop_rp_v	0.9
sweep_points	11
sweep_repeats	1
sweep_settle_s	1.0
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
phi1_v_lambda	12.2
phi2_v_lambda	12.2
phi1_actuator_volts_per_rp_volt	16.875
phi2_actuator_volts_per_rp_volt	16.875
rp_output_min_voltage	0.0
rp_output_max_voltage	1.0